

Anticipating numeric conflicts

Fast planning under resource constraints

Robots need useful plans. Quickly.

Planning time is part of the task.

Fast guidance can miss resource interactions.

Planner / heuristic	Valid	Total time (s)	Objective gap*
Metric-FF / numeric-hff	30/30	8.6	8.2%
Panino / hadd-novelty	30/30	397.8	35.0%
Count Downward / irhff	28/30	626.8	18.1%
NFD / irhadd	27/30	1,407.7	7.2%
ENHSP / hadd	25/30	1,731.2	5.8%
ENHSP / hradd	25/30	1,596.1	12.0%

*Median excess over observed best, on solved tasks; solved sets differ. Total time includes timeouts.

A domain name is not a difficulty model.

Domain	p000	p001	p002	p003	p004
Blocksworld	97.7%	95.3%	88.4%	83.7%	81.4%
Logistics	95.3%	62.8%	32.6%	18.6%	16.3%
Books	93.3%	75.6%	44.4%	17.8%	15.6%
Watering	91.1%	60.0%	53.3%	40.0%	11.1%
Barman	78.4%	32.4%	8.1%	10.8%	8.1%
Assembly	93.3%	73.3%	66.7%	51.1%	22.2%

VAL-valid rate across tested configurations; technical incompatibilities excluded.

Discussion informed by our benchmark and resource-control experiments

CONSTRAINTS

Tighter constraints can help search.

Logistics / configuration	L/L states	T/T states
p003 · Count Downward / irhff	737	286
p003 · ENHSP / hadd	202	45,992
p003 · NFD / irhadd	217	32,946
p004 · Count Downward / irhff	131,514	1,541

Expanded states. L/L: loose fuel + budget; T/T: tight fuel + budget. Same problem structure within each pair.

DELAYED CONFLICT

A feasible action can lead to an infeasible future.

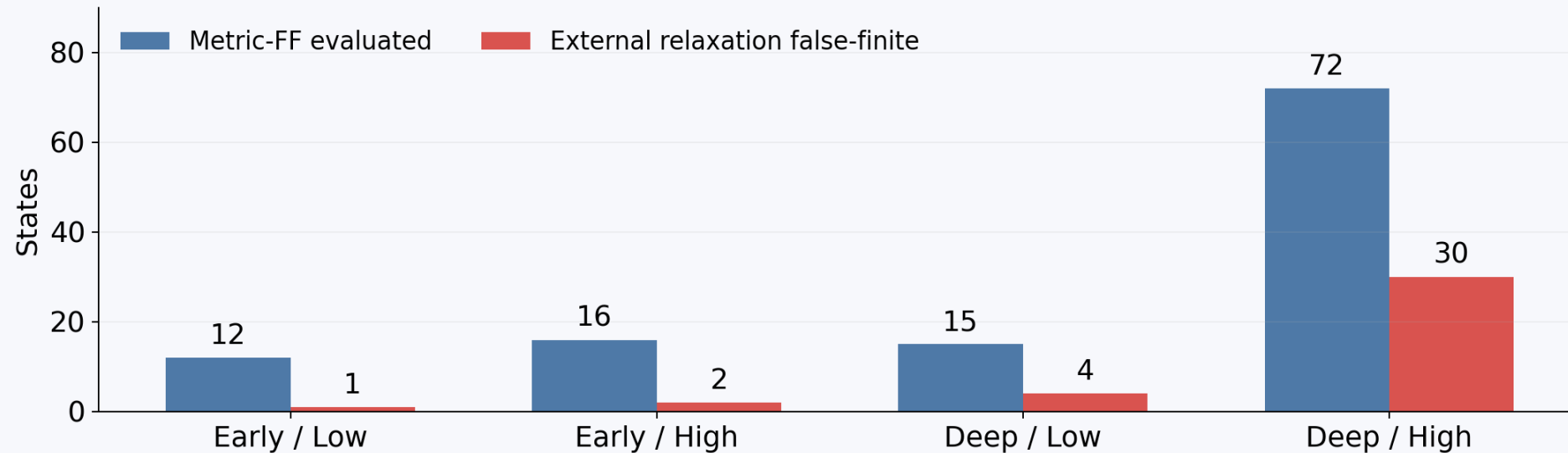
Barman p001	Blocked-zero (s)	One-short (s)
Metric-FF / numeric-hff	0.20	1.41
Count Downward / irhff	0.20	2.61
NFD / irhadd	1.00	32.32
ENHSP / hadd	0.40	5.62
ENHSP / hradd	0.40	5.62

Wall time to unsolved termination. Both variants unsolvable; deficit magnitude also differs.

Discussion informed by our Barman stock experiments

WORKING HYPOTHESIS

Misleading choices can survive too long.



Synthetic pilot. False-finite labels use an external relaxation, not instrumented Metric-FF values.

Hypothesis motivated by synthetic and domain-schema micro experiments

HEURISTIC DEPENDENCE

**An extra branch matters
only if search follows it.**

Planner / heuristic	Watering E/L	E/H	D/L	D/H
Metric-FF / numeric-hff	24	44	34	64
Count Downward / irhff	5	5	12	12
NFD / irhadd	18	64	49	153
ENHSP / hadd	5	5	12	12
ENHSP / hradd	5	5	12	12

E/D: early/deep; L/H: low/high branching. Metric-FF: evaluated; others: expanded. Unsolvable micro tasks.

Discussion informed by our Watering and Logistics micro experiments

Bring future resource conflicts into current search guidance.

The signal must be useful enough—and cheap enough.

Richer numeric reasoning already exists.

The question is when its extra computation pays off.

Selective evaluation already has a foundation.

Our candidate signal: where numeric relaxation may mislead.

Use a cheap signal to allocate additional reasoning.

Conflict signal → Selective evaluation → Search guidance

Potential-based signals and semantic priors remain candidate designs.

Where does additional reasoning actually help?

Compare heuristic decisions on the same states.

Next step: state-level solvability, heuristic values and evaluation cost

Can we predict misleading guidance before the conflict?

Can that prediction save more time than it costs?

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